ENTERPRISE PLATFORM SPECIFICATION

RLINS — Rental Lease Invoice Notification System

Automated Invoicing, Multi-Page PDF Splitting, Oracle ERP Integration & Jenkins CI/CD

SUMMARY 1. Executive Summary (Short & Crisp Version)

RLINS (Rental Lease Invoice Notification System) is an automated enterprise lease billing and tenant notification web application that splits consolidated monthly bill PDFs, intelligently pairs each receipt with tenants using regex pattern matching, and dispatches them via an SMTP email engine with real-time Server-Sent Events (SSE) progress tracking. Built with **Node.js, Express, SQLite (WAL mode), and Vanilla JavaScript/CSS**, it features a comprehensive tenant management dashboard with Excel import/export, role-based session security, and a real-time two-way synchronization bridge to an enterprise **Oracle Database via SQL*Plus**. The application is containerized with **Docker & Docker Compose** for local intranet distribution on port 5178 and fully automated via a **Jenkins CI/CD Pipeline using GitHub SCM Polling** for zero-downtime, continuous deployments.

IN-DEPTH 2. Comprehensive Architecture & Functional Overview (Detailed Version)

RLINS (Rental Lease Invoice Notification System / Kiraidaar) is an enterprise-grade lease billing automation and tenant communication platform designed to streamline monthly estate bill processing, tenant reconciliation, and automated email dispatching. The core workflow begins by taking large, consolidated monthly lease bill PDFs, splitting them page-by-page using automated PDF extraction engines (`split_pdf_by_page.js`), and executing an intelligent regex-based matching algorithm (`backend/matchingEngine.js`) that scans filenames, customer codes, and bill metadata to automatically pair each receipt with its corresponding tenant with 100% confidence. The system provides a centralized web dashboard featuring real-time billing metrics, full tenant roster lifecycle management (CRUD operations with duplicate-code safeguards), Excel/CSV bulk import/export capabilities (`xlsx`), in-browser PDF previewing, and a manual rematching tool for ambiguous bills. For corporate integration, RLINS implements an asynchronous two-way bridge (`backend/oracleSync.js`, `host_bridge.js`) that interfaces directly with an enterprise **Oracle Database** (`PAYTEMP.LEASED_CUSTOMERS_DTLS` and `LEASED_SETTINGS_DTLS`) via native SQL*Plus, syncing tenant directories, template configurations, and deletion cascades in real time without blocking the UI. Bills are distributed via an email engine (`backend/emailService.js`) powered by **Nodemailer**, supporting SMTP/TLS, dynamic variable templating, a safe simulation/dry-run mode, and live progress streaming to the browser via **Server-Sent Events (SSE)**. The entire solution is hardened with token-based session security, 24-hour TTL expiration, and containerized using **Docker and Docker Compose** bound to `0.0.0.0:5178` for company-wide intranet access, with persistent host storage for its SQLite database (`kiraidaar.db`) and PDF volumes. The lifecycle is automated using a declarative **Jenkins CI/CD Pipeline** (`Jenkinsfile`) integrated with **GitHub SCM Polling** (`H/2 * * * *`), enabling automated syntax validation, image rebuilds, zero-downtime container recreation, and automatic health-endpoint verification on every commit.

📄 Automated PDF Decomposition

Separates massive multi-page billing documents into isolated single-page PDFs per tenant with zero layout corruption.

🎯 Intelligent Regex Matching Engine

Extracts party codes, names, and invoice IDs with 100% automated pairing and interactive manual override tools.

📊 Roster & Excel Spreadsheet Suite

Full CRUD tenant roster with duplicate prevention, master roster export, and pre-formatted XLSX sample template downloads.

🔄 Oracle ERP Live Bidirectional Sync

Non-blocking background sync with Oracle PAYTEMP tables via native SQL*Plus bridge with real-time deletion cascades.

✉️ SMTP Dispatcher & Live SSE Stream

Batch email distribution with template variable injection, dry-run simulation mode, and live browser progress updates.

🚀 Zero-Downtime Jenkins CI/CD

Automated GitHub SCM Polling (2-min interval), layer-cached Docker builds, and zero-loss persistent SQLite volume mounting.

STACK 3. Complete Technology Stack Matrix

Layer / Domain	Technologies & Libraries	Architectural Role & Implementation
Frontend Client	HTML5, CSS3 Glassmorphism, Vanilla ES6+	Responsive dashboard, custom animated modals, real-time SSE stream listeners, in-browser PDF previewer.
Backend REST API	Node.js (v22), Express.js 5.x	REST endpoints, session token auth with 24h TTL, non-blocking async architecture, child_process execution.
Local Persistence	SQLite3 (WAL Mode, kiraidaar.db)	High-concurrency ACID local database, PRAGMA busy_timeout=10000, persistent bind mount to host D:/RLNS/data.
Enterprise ERP	Oracle DB 10g/11g / SQL*Plus Bridge	Two-way sync for PAYTEMP.LEASED_CUSTOMERS_DTLS & LEASED_SETTINGS_DTLS via host_bridge.js on port 3005.
Document Engine	pdf-lib, pdf-parse, xlsx (SheetJS), Multer	Multi-page bill decomposition, PDF text parsing, Excel template generation, and multipart upload handling.
Notification	Nodemailer (SMTP / TLS / Queue)	Automated bill delivery, live Server-Sent Events progress reporting, simulation/dry-run mode, and audit logs.
Containerization	Docker & Docker Compose	Multi-stage Node 22 slim container, host-gateway networking, 0.0.0.0:5178 intranet binding, built-in healthcheck.
CI/CD Automation	Jenkins Pipeline & GitHub SCM Polling	Declarative Jenkinsfile, automated syntax validation, smart layer caching, and zero-downtime redeployment.